Scalability & Future Plan

Date	10 July 2026
Team ID	XXXXXX
Project Name	Smart Lender Loan Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only local deployment using the Flask development serve	Limits access to users on the local machine	High
2	Uses a static dataset for training	Prediction accuracy depends on historical data and is not updated automatically	Medium
3	No user authentication or database integration	User data and prediction history cannot be stored	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of local users	Deploy on cloud platforms such as Render, AWS, or Azure with load balancing
2	Data Storage	Uses CSV dataset and trained model files	Integrate MySQL or MongoDB for secure and scalable data storage
3	Performance	Prediction is performed using a locally hosted model	Optimize the ML model and use cloud-based inference for faster predictions
4	Security	Basic input validation is implemented	Add user authentication, role-based access control, HTTPS, and data encryption

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate MySQL/MongoDB and user authentication	2–3 Months	Secure data storage and user management
Phase 3	Deploy the application on a cloud platform (Render/AWS/Azure)	3–4 Months	Improved accessibility, scalability, and availability
Phase 4	Add advanced Machine Learning models, loan history tracking, analytics dashboard, and banking API integration	4–6 Months	Higher prediction accuracy, enhanced functionality, and better decision support